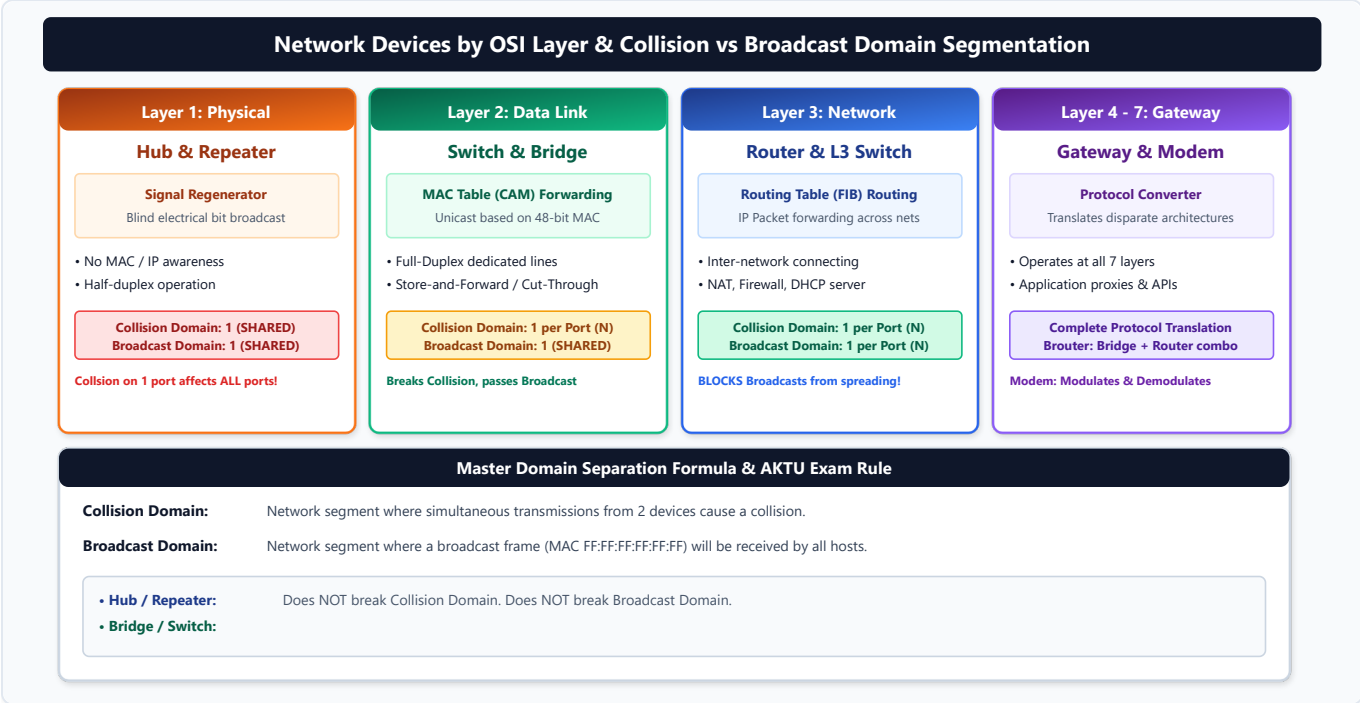


Module 14: Network Devices, Components & Unit 1 AKTU PYQs

AKTU BCS603 Grand Finale: Complete hardware device taxonomy, Collision vs Broadcast domain segmentation rules, and model solutions for AKTU PYQs.



BREAKS Collision Domain (1 domain per port). Does NOT break Broadcast Domain. • Router / Gateway: **BREAKS Collision Domain AND BREAKS Broadcast Domain** (Stops broadcasts cold!).

1. Domain Segmentation Formula (Quick Revision)

Connecting Device	Number of Ports	Collision Domains Created	Broadcast Domains Created
Repeater / Hub	N	1 (Single shared collision pool)	1 (Single broadcast pool)
Bridge / Switch	N	N (1 independent domain per port)	1 (Floods broadcast frames)
Router / Gateway	N	N (1 independent domain per port)	N (Filters/blocks broadcasts)

2. Bridge Learning & Filtering Algorithm

Bridge incoming frames ke source MAC address se network topology learn karta hai:

Frame Arrives on Port P with [Src: MAC_S, Dst: MAC_D]

- Update Table: MAC_S -> Port P
- If MAC_D is in Table:

If Port(MAC_D) == P: DROP / FILTER Frame (Local traffic)

Else: FORWARD Frame to Port(MAC_D)
- Else: FLOOD Frame to all ports except P

AKTU 10-Mark Success Blueprint: Hardware devices question me har device ka OSI layer, working principle (CAM table / Routing table), aur Collision/Broadcast domain separation status zaroor mention karein!